

Origin of the CEEX–PHOKHARA discrepancy in the pion angular distributions

KLOE large-angle scenario, $e^+e^- \rightarrow \pi^+\pi^-\gamma$ at NLO, $F_\pi = 1$, no vacuum polarisation

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Abstract

In the KLOE large-angle (KLOE-LA) scenario the NLO predictions of CEEX and PHOKHARA 10.0 for $e^+e^- \rightarrow \pi^+\pi^-\gamma$ agree in the total cross section and in every charge-even distribution, but disagree in the pion polar angles: the forward–backward asymmetry of the π^+ is $A(\theta^+) = 0.2375$ in CEEX (and 0.2372 in BabaYaga) against 0.2787 in PHOKHARA. We show that the difference is entirely due to two errors in the one-photon ISR×FSR interference at NLO in PHOKHARA (routine `ampL0`): a misplaced parenthesis that adds three helicity terms of the tree interference a second time, and a block that keeps the tree interference together with an analytic ISR soft function that the calculation replaces by the eikonal elsewhere. The second error makes the PHOKHARA asymmetry depend on the unphysical soft-photon cutoff. A point-by-point comparison of the one-photon matrix elements shows that the two errors account for the whole charge-odd difference to 2.5×10^{-5} of the tree. With both corrected, PHOKHARA gives $A(\theta^+) = 0.23724(10)$ averaged over the cutoff and 0.23725(3) with 5×10^9 events, independent of the cutoff; CEEX (0.23726(3)) and BabaYaga (0.23724) agree with it to 3×10^{-5} .

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*Analysis carried out with Claude Code (Anthropic). Repository: https://github.com/jpaltrinieri/CEEX_PHOKHARA_ANALYSIS

1 Summary of findings

1. **The disagreement is reproduced** with a fresh PHOKHARA build and fresh CEEX runs (Section 4): totals agree to 10^{-4} , charge-even shapes agree, and only $\theta^\pm$ and θ_{avg} differ ($\chi^2/\text{ndf} \approx 10^3$). At LO the two codes agree exactly.
2. **The PHOKHARA asymmetry depends on the soft cutoff w** : $A(\theta^+)$ changes by $+0.0156$ per decade of w while σ is constant (Section 6.2).
3. **The whole effect sits in the charge-odd part of the one-photon contribution** (Born + virtual + soft); the hard two-photon emission agrees between the codes at a matched cutoff.
4. **Two errors in PHOKHARA explain it point by point** (Section 6):
 - **H1** – `phokhara_10.0.f:9969-9982`: the factor $(v_f + v_s)$ multiplies only the last of four helicity terms of the tree interference;
 - **H2** – block B5, the ISR one-loop current times the FSR tree (`phokhara_10.0.f:9998-10007`; `ISRcurrent_no_mass_terms.f:134,210`): an extra $\text{Re}[F^*I](1 + S_{\text{ISR}}(w))$.

Subtracting both leaves a per-point residual of 2.5×10^{-5} of the tree.

5. **Both errors are in the official PHOKHARA 10.0 release** (October 2020) and came with the full-NLO pion mode, which is new in that version (Section 6.3).
6. **With both corrected**, $A(\theta^+)$ no longer depends on w and agrees with CEEX and BabaYaga, in total and separately in the one- and two-photon parts (Section 7). Totals and charge-even distributions are unchanged.

2 Setup

Scenario. KLOE-LA (Strong2020 KLOE-I), $\sqrt{s} = 1.02$ GeV: $50^\circ \leq \theta_\pm \leq 130^\circ$; $|p_\pm^z| > 90$ MeV or $p_\pm^\perp > 160$ MeV; at least one photon with $50^\circ \leq \theta_\gamma \leq 130^\circ$ and $E_\gamma > 20$ MeV; $0.1 \leq M_{\pi\pi}^2 \leq 0.85$ GeV². Angles are measured from the electron beam in both codes. Histograms: the 18 KLOE-I observables with 600 uniform bins. The forward–backward asymmetry is $A = [\sigma(\theta < 90^\circ) - \sigma(\theta > 90^\circ)]/\sigma$.

Codes.

- **CEEX-main** (commit e05d007): `pipig -scenario=KLOE-LA -NLO`, one-loop amplitudes from GoSam (`eepipig_noVP_1L_onshell`, `eepipigg`), YFS soft subtraction with cutoff $E_{\text{min}} = 10^{-5}$ GeV (10^{-4} GeV only in the cutoff checks); 5 MPI ranks per task.
- **PHOKHARA 10.0** (`PHOKARA_10.0 @ 133d68b`) with the KLOE-I cut file, card `nlo=nlo2=1`, `IFSNLO=1`, `ISR+INT+FSR`, `ivac=0`, `FF_pion=-1` ($F_\pi = 1$), no GVMD, no f_0 , soft cutoff $w = 10^{-5}$ (soft region $E_\gamma < w\sqrt{s}$). The production binary `phokhara_nostop` only disables the abort on “Max(i) too small”; only weighted events are used.
- **BabaYaga** and the earlier PHOKHARA reference set (5 Aug) are taken from the Strong2020 comparison files.

Pipeline. SLURM arrays per run name; CEEX rank outputs are screened with `check_total.sh` and outliers are moved aside (`quarantine.sh`) before merging; `plot_compare.py` produces the distribution and ratio plots. Everything is in the repository https://github.com/jpaltrinieri/CEEX_PHOKHARA_ANALYSIS.

3 How PHOKHARA computes $\pi^+\pi^-\gamma$ at NLO

This section describes what PHOKHARA 10.0 computes for $\pi^+\pi^-\gamma$ at full NLO, the calculation of this report ($F_\pi = 1$, no vacuum polarisation): first the overall structure, then the routines, the UV and IR treatment, and finally the two blocks where the errors are (Section 3.5). Line numbers refer to the unpatched `phokhara_10.0.f` unless another file is named. Statements marked (*inferred*) are our reading of the code; the code itself does not document them.

3.1 Overall structure

Amplitudes. At tree level the one-photon amplitude is $M = I + F$:

- I (ISR): the photon is emitted from the $e^\pm$ line, and $\gamma^*(q^2) \rightarrow \pi\pi$ with the current $J^\mu = (q_+ - q_-)^\mu F_\pi(q^2)$;
- F (FSR): $\gamma^*(s) \rightarrow \pi\pi\gamma$ in scalar QED, i.e. emission from π^+ , from π^- and the seagull (point-like pions).

$|I|^2$ and $|F|^2$ are charge-even; the interference $2\text{Re}[I^*F]$ is charge-odd (odd under $\pi^+ \leftrightarrow \pi^-$) and is the only source of the pion forward-backward asymmetry. At NLO every one-loop and soft term is either charge-even or charge-odd; Table 1 lists them all.

Event classes. Two parameters split the photon phase space:

- w , the soft-photon cutoff: a photon with $E_\gamma < w\sqrt{s}$ (CM frame) is soft, is never generated, and is integrated analytically;
- `gmin`, the minimum energy of the tagged photon.

Each event is a one-photon or a two-photon event; one random number picks the class with probability proportional to its maximum weight from the scan pass.

- **One-photon events** (one hard photon above `gmin`) carry in a single weight the tree $|M|^2$, all one-loop virtual corrections, and the integrated soft emission.
- **Two-photon events** carry the exact tree $|M|^2$ for $e^+e^- \rightarrow \pi^+\pi^-\gamma\gamma$: one photon is tagged ($E \geq \text{gmin}$, inside the angular cuts), the other only has to be above $w\sqrt{s}$.

The two classes add up to the full $\mathcal{O}(\alpha)$ correction, and the w dependence must cancel in the sum. This is phase-space slicing; there are no local subtraction terms.

Precision. The one-photon kinematics are generated in quad precision (`momenta_quad.f`). Only `virt_pions` uses the quad momenta; all other pieces use rounded double copies.

3.2 The one-photon weight

Notation. Write the one-photon amplitude to $\mathcal{O}(e^5)$ as

$$M = \underbrace{I + F}_{\text{tree}} + \underbrace{\delta I_e + \delta I_\pi + \delta F_e + \delta F_\pi + \delta B_e + \delta B_\pi}_{\text{one loop}},$$

where the one-loop amplitudes are

- δI_e : ISR with the loop on the electron line (vertex and self-energy corrections around the emitted photon);
- δI_π : ISR with the $\gamma^*\pi\pi$ vertex correction at q^2 ;
- δF_e : FSR with the $e^+e^-\gamma^*$ vertex correction at s ;
- δF_π : FSR with the loop on the pion line (self-energies, vertices, seagull);

- $\delta B_e, \delta B_\pi$: two-photon exchange between the electron and pion lines (boxes and pentagons), with the real photon on the electron or on the pion line.

Figure 1 shows one example diagram for each class.

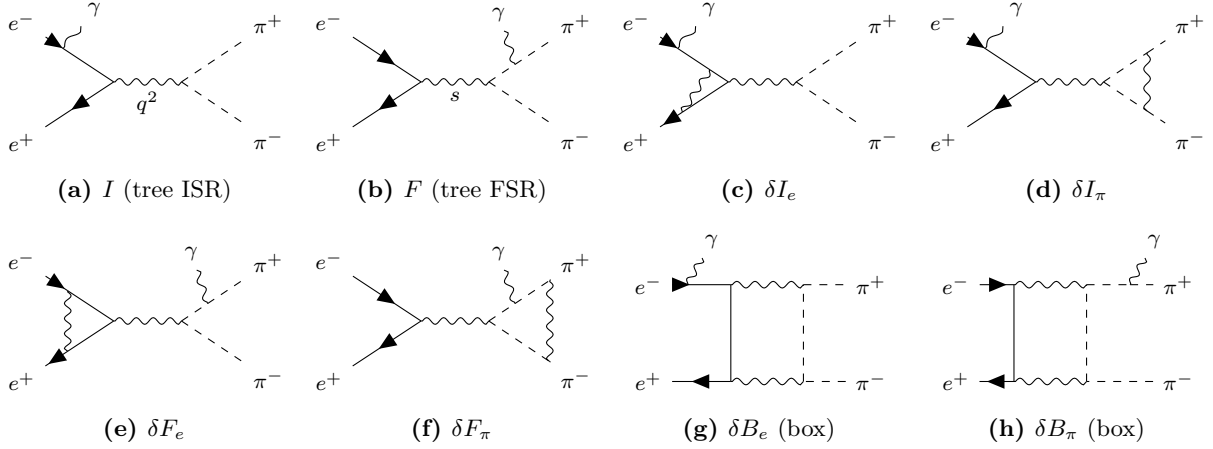


Figure 1: One example diagram per amplitude class of Section 3.2, with the real photon γ ($E_\gamma > w\sqrt{s}$). Top: the tree amplitudes I and F and the ISR corrections; bottom: the FSR corrections and the two-photon exchange. Each class also contains the diagrams with the photon on the other leg, the self-energies, the crossings ($\pi^+ \leftrightarrow \pi^-$, for δB the crossed photon lines) and, for F and δF_π , the seagull $\gamma\gamma\pi\pi$ vertex. The δB examples are boxes (real photon on an external leg); with the real photon on an internal line of the loop they become the pentagons.

The two vertex corrections factorise: $\delta F_e = \frac{1}{2}v_s F$ and $\delta I_\pi = \frac{1}{2}v_f I$, with v_s (`ver_s`) and v_f (`ver_f=vertL0`) real $|M|^2$ -level factors.

From the amplitude expansion to the routines. Squaring M and keeping $\mathcal{O}(\alpha)$ relative to the tree, every term is an interference $2\text{Re}[X^*Y]$ of a tree amplitude $X \in \{I, F\}$ with a tree or one-loop amplitude Y . PHOKHARA computes them in four routines:

$$\begin{aligned}
 |M|^2 = & \underbrace{|I|^2 + 2\text{Re}[I^*\delta I_e]}_{\text{amplit}} \\
 & + \underbrace{|F|^2 + 2\text{Re}[F^*\delta F_e]}_{\text{B1}} + \underbrace{2\text{Re}[I^*\delta I_\pi]}_{\text{B2}} + \underbrace{2\text{Re}[I^*F]}_{\text{B3}} + \underbrace{2\text{Re}[I^*\delta F_e + F^*\delta I_\pi]}_{\text{B4}} + \underbrace{2\text{Re}[F^*\delta I_e]}_{\text{B5}} \\
 & \underbrace{\hspace{10em}}_{\text{helicityampL0}} \\
 & + \underbrace{2\text{Re}[(I+F)^*(\delta F_\pi + \delta B_e + \delta B_\pi)]}_{\text{virt_pions}} + \underbrace{|I+F|^2 \mathcal{S}(w)}_{\text{NLOpions}}, \tag{1}
 \end{aligned}$$

where $\mathcal{S}(w)$ is the eikonal factor for soft emission below $w\sqrt{s}$. The same assignment as a grid, one cell per $2\text{Re}[X^*Y]$ (the F^*I cell is the same term as I^*F):

	I	F	δI_e	δI_π	δF_e	δF_π	$\delta B_e, \delta B_\pi$
I^*	amplit	B3	amplit	B2	B4	virt_pions	virt_pions
F^*	(B3)	B1	B5	B4	B1	virt_pions	virt_pions

The pattern follows from what each routine can do:

- `amplit` is a tensor contraction $L^{\mu\nu}H_{\mu\nu}$, which only exists for $|\text{ISR}|^2$: it gives the I^* entries with I and δI_e , and nothing else.
- The vertex corrections factorise, $\delta F_e = \frac{1}{2}v_s F$ and $\delta I_\pi = \frac{1}{2}v_f I$, so every cell with them is a tree helicity sum times v_s or v_f . Together with the tree $|F|^2$ and $2\text{Re}[I^*F]$ they are all in `helicityampL0` (B1–B4).

- δI_e appears twice. Against I^* it is in **amplit** (analytic, m_e kept); against F^* it needs the amplitude, which **helicityampL0** takes from **NL0_ISR_no_mass_terms** (massless electrons) in B5; the effect of the electron mass is discussed in Appendix C.
- The non-factorising loops δF_π , δB_e , δB_π are genuine one-loop amplitudes, computed in **virt_pions** against the full tree $I + F$.

Table 1 lists the same terms with their charge factors. Counting charges ($I \propto Q_e^2 Q_\pi$, $F \propto Q_e Q_\pi^2$, each loop adds two charges), a term is charge-odd when it has an odd power of Q_π . The charge-odd cells are I^*F , $I^*\delta F_e$, $F^*\delta I_\pi$, $F^*\delta I_e$, $I^*\delta F_\pi$, $I^*\delta B_e$ and $F^*\delta B_\pi$, plus the odd part of the soft term.

Table 1: Every contribution to the one-photon weight and where PHOKHARA computes it. B1–B5 are the blocks of **ampL0** (Table 2). $S_{\text{ISR}}(w)$ is the analytic ISR soft function of Section 3.4. The hard two-photon emission, the other half of the real correction, is computed separately in **gen_2ph**.

Contribution	Charges	C	Computed in	Remark
<i>Tree</i>				
$ I ^2$	$Q_e^4 Q_\pi^2$	+	amplit	tensor $L^{\mu\nu} H_{\mu\nu}$
$ F ^2$	$Q_e^2 Q_\pi^4$	+	ampL0 B1	
$2\text{Re}[I^*F]$	$Q_e^3 Q_\pi^3$	–	ampL0 B3	
<i>One loop</i>				
$2\text{Re}[I^*\delta I_e]$	$Q_e^6 Q_\pi^2$	+	amplit	Rodrigo, m_e kept; $-S_{\text{ISR}} I ^2$
$2\text{Re}[I^*\delta I_\pi] = v_f I ^2$	$Q_e^4 Q_\pi^4$	+	ampL0 B2	
$2\text{Re}[F^*\delta F_e] = v_s F ^2$	$Q_e^4 Q_\pi^4$	+	ampL0 B1	
$2\text{Re}[F^*\delta F_\pi]$	$Q_e^2 Q_\pi^6$	+	virt_pions	virtual_FSRHelSum
$2\text{Re}[I^*\delta F_e + F^*\delta I_\pi]$ $= \text{Re}[I^*F](v_s + v_f)$	$Q_e^5 Q_\pi^3, Q_e^3 Q_\pi^5$	–	ampL0 B4	H1
$2\text{Re}[F^*\delta I_e]$	$Q_e^5 Q_\pi^3$	–	ampL0 B5	massless e ; H2
$2\text{Re}[I^*\delta F_\pi]$	$Q_e^3 Q_\pi^5$	–	virt_pions	virtual_FSRHelSum
$2\text{Re}[I^*\delta B_e]$	$Q_e^5 Q_\pi^3$	–	virt_pions	Pentabox_full_isr , cross_ISR
$2\text{Re}[F^*\delta B_e]$	$Q_e^4 Q_\pi^4$	+	virt_pions	same routines
$2\text{Re}[I^*\delta B_\pi]$	$Q_e^4 Q_\pi^4$	+	virt_pions	Pentabox_PIFSR , cross_FSR
$2\text{Re}[F^*\delta B_\pi]$	$Q_e^3 Q_\pi^5$	–	virt_pions	same routines
<i>Soft emission, $E_\gamma < w\sqrt{s}$</i>				
$ M_{\text{tree}} ^2 \times (ee, \pi\pi \text{ eikonal})$	$\times Q_e^2, \times Q_\pi^2$	$\pm$	NLOptions	odd part from $2\text{Re}[I^*F]$
$ M_{\text{tree}} ^2 \times (e\pi \text{ eikonal})$	$\times Q_e Q_\pi$	$\mp$	NLOptions	odd part from $ I ^2 + F ^2$

amplit: first line of Equation (1) (lines 2069, 6757–7037). Starting from the definition:

$$\begin{aligned}
\text{amplit} &= L^{\mu\nu} H_{\mu\nu} && \text{definition: leptonic} \times \text{hadronic tensor} \\
&= |I|^2 + (2\text{Re}[I^*\delta I_e])_{\text{V+S}} && \text{Bornvirtualsoft: LO + virtual + analytic soft} \\
&= |I|^2 + 2\text{Re}[I^*\delta I_e]_{\text{V}} + S_{\text{ISR}}(w) |I|^2 && \text{split V+S into virtual and soft} \\
&\longrightarrow |I|^2 + 2\text{Re}[I^*\delta I_e]_{\text{V}} && \text{subtract } S_{\text{ISR}}(w) |I|^2 \text{ (lines 6838–6856)}
\end{aligned}$$

The ingredients:

- $H^{\mu\nu} \propto |F_\pi(q^2)|^2 (q_+ - q_-)^\mu (q_+ - q_-)^\nu$ (**HadronicTensorISR**): the $\gamma^* \rightarrow \pi\pi$ vertex, squared.
- $L^{\mu\nu}$ (**LeptonicTensor1ISR**): the electron line with the emitted photon, squared and summed over the photon polarisation, $L = L_{\text{LO}} + \frac{\alpha}{\pi} L_{\text{NLO}}$. The coefficients come from **Bornvirtualsoft** (G. Rodrigo et al., EPJC 22 (2001) 81; J.H. Kühn, G. Rodrigo, EPJC 25 (2002) 215), with m_e kept. $L_{\text{LO}} H = |I|^2$, and $\frac{\alpha}{\pi} L_{\text{NLO}} H$ is the IR-finite sum of the one-loop ISR correction and the analytic soft emission below $w\sqrt{s}$.
- V: the virtual correction with its IR pole dropped (dimensional regularisation, $\mu^2 = s$), the convention of the eikonal integrals in **NLOptions**; (*inferred*) from the form of S_{ISR} .

- $S_{\text{ISR}}(w)$: the analytic ISR soft function (Section 3.4). Once it is subtracted, the soft emission it stood for is supplied by **NLOpions**.

amplit contains nothing else: no FSR, no interference, and not the pion-vertex correction (that is B2).

helicityampL0 $\rightarrow$ **ampL0**: **second line of Equation (1)** (lines 9687–10021). Starting from the definition,

$$\text{helicityampL0} = \frac{(4\pi\alpha)^3}{4} \text{ampL0} \times (\text{phase space}), \quad \text{ampL0} = \sum_{\lambda_\gamma=1,2} (B_1 + B_2 + B_3 + B_4 + B_5),$$

where the $1/4$ is the $e^\pm$ spin average and B1–B5 are helicity sums $\langle X, Y \rangle$ of tree and one-loop amplitudes (defined below, Table 2). Per photon polarisation:

$$\sum_i B_i = |F|^2(1 + v_s) + |I|^2 v_f + 2\text{Re}\langle I, F \rangle + \text{Re}\langle I, F \rangle(v_f + v_s) + \text{Re}\langle F, I_{1\text{L}} \rangle + \Delta B_4$$

$\downarrow$ *factorised vertex corrections*, $\delta F_e = \frac{1}{2}v_s F$ and $\delta I_\pi = \frac{1}{2}v_f I$

$$= |F|^2 + 2\text{Re}[F^* \delta F_e] + 2\text{Re}[I^* \delta I_\pi] + 2\text{Re}[I^* F] + 2\text{Re}[I^* \delta F_e + F^* \delta I_\pi] + \text{Re}\langle F, I_{1\text{L}} \rangle + \Delta B_4$$

$\downarrow$ *one-loop ISR amplitude of B5*, $I_{1\text{L}} = I(1 + S_{\text{ISR}}) + 2\delta I_e$, so $\text{Re}\langle F, I_{1\text{L}} \rangle = 2\text{Re}[F^* \delta I_e] + \Delta B_5$

$$= [\text{second line of Equation (1)}] + \Delta B_4 + \Delta B_5$$

$\downarrow$ *the fix (H1, H2) removes ΔB_4 and ΔB_5*

$$\rightarrow [\text{second line of Equation (1)}].$$

The first line is the code block by block (Table 2), with B4 written as its intended form plus the remainder ΔB_4 left by the misplaced parenthesis. The two extra terms are both proportional to the tree interference:

$$\Delta B_4 = \text{Re} \sum_{k \leq 3} T_k(I)^* T_k(F) (1 - (v_f + v_s)), \quad \Delta B_5 = \text{Re}\langle I, F \rangle (1 + S_{\text{ISR}}(w)).$$

The decomposition of $I_{1\text{L}}$ is read from **V1(1)** (Section 6.2) and confirmed numerically by a per-point comparison with CEEX at 988 one-photon points.

The ingredients:

- Amplitudes as 2×2 Weyl blocks A and B, with **plus**(v) = $v^0 - \vec{v} \cdot \vec{\sigma}$ and **minus**(v) = $v^0 + \vec{v} \cdot \vec{\sigma}$:

Amplitude	A block	B block
I (tree ISR, from blocksL0)	ma	mb
F (tree FSR, from ddvec)	ddmi	ddpl
$I_{1\text{L}}$ (ISR at one loop, from NLO_ISR_no_mass_terms)	ami	apl

- Helicity amplitudes: the beam spinors enter through four fixed combinations of the blocks (lines 9939–9942), with $e_b = E + |\vec{p}|$ of the positron,

$$\begin{aligned} T_1(X) &= m_e(B_{11} - A_{11}), & T_3(X) &= -e_b A_{21} + \frac{m_e^2}{e_b} B_{21}, \\ T_2(X) &= m_e(B_{22} - A_{22}), & T_4(X) &= e_b B_{12} - \frac{m_e^2}{e_b} A_{12}. \end{aligned}$$

(*Inferred*) T_1, T_2 are the m_e -suppressed helicity-flip amplitudes and T_3, T_4 the two leading helicity-conserving ones.

- Helicity sums: $\langle X, Y \rangle \equiv \sum_{k=1}^4 T_k(X)^* T_k(Y)$ and $|X|^2 \equiv \langle X, X \rangle$.
- v_s (**ver_s**) and v_f (**vertL0**): the electron-vertex correction at s and the pion-vertex correction at q^2 , analytic, with their soft parts removed (Section 3.4).

Table 2: The blocks of `ampL0`, per photon polarisation. Vacuum-polarisation factors `cvac` are 1 in our setup and omitted.

Block	Code computes	Physics content	Lines
B1	$ F ^2(1 + v_s)$	$ F ^2 + 2\text{Re}[F^*\delta F_e]$	9938–9943
B2	$ I ^2 v_f$	$2\text{Re}[I^*\delta I_\pi]$	9945–9951
B3	$2\text{Re}\langle I, F \rangle$	$2\text{Re}[I^*F]$	9954–9966
B4	$\text{Re}\sum_{k\leq 3} T_k(I)^*T_k(F) + \text{Re}[T_4(I)^*T_4(F)](v_f + v_s)$	intended $\text{Re}\langle I, F \rangle(v_f + v_s)$	9969–9982 H1
B5	$\text{Re}\langle F, \bar{I}_{1L} \rangle$, with $I_{1L} = I(1 + S_{\text{ISR}}) + 2\delta I_e$	intended $2\text{Re}[F^*\delta I_e]$	9984–10007 H2

virt_pions: third line of Equation (1) (`virtualpions.f`, lines 74–220). Starting from the definition,

$$\frac{\text{virt_pions}}{\text{phase space}} = \frac{(4\pi\alpha)^3}{4} \frac{\alpha}{4\pi} 2\text{Re}[(P_\pi + P_\pi^{\text{cr}})F_s^* + (P_e + P_e^{\text{cr}})F_q^* + A_{\text{GVMD}}] \\ + 32\pi^2\alpha^4 \text{Re}[V_\pi] F_s^*$$

↓ $F_s = F_\pi(s)$, $F_q = F_\pi(q^2)$ are 1 ($F_\pi = 1$, no VP); $A_{\text{GVMD}} = 0$ (no GVMD)

$$= \frac{(4\pi\alpha)^3}{4} \frac{\alpha}{4\pi} 2\text{Re}[P_\pi + P_\pi^{\text{cr}} + P_e + P_e^{\text{cr}}] + 32\pi^2\alpha^4 \text{Re}[V_\pi]$$

↓ each routine returns its loop amplitude summed over helicities against the tree helicity vector `hel()`, which contains both I and F

$$= 2\text{Re}[(I + F)^*\delta B_\pi] + 2\text{Re}[(I + F)^*\delta B_e] + 2\text{Re}[(I + F)^*\delta F_\pi] \\ = [\text{third line of Equation (1)}].$$

The ingredients (diagram content inferred from the propagator masses):

- P_π , P_π^{cr} (`Pentabox_PIFSRHelSum`, `cross_FSRHelSum`): δB_π , two-photon exchange with the real photon on the pion line (pentagon masses $(m_e, 0, m_\pi, m_\pi, 0)$, plus the boxes), and its $\pi^+ \leftrightarrow \pi^-$ crossing.
- P_e , P_e^{cr} (`Pentabox_full_isrHelSum`, `cross_ISRHelSum`): δB_e , the same with the real photon on the electron line (masses $(m_e, m_e, 0, m_\pi, 0)$), and its crossing.
- V_π (`virtual_FSRHelSum` and `...cr`, `test_vFSR.f`): δF_π , the pion-line loops, only massless and m_π propagators, including the on-shell counterterms (Section 3.3); double precision, QCDLoop integrals.
- All loops: quad precision (except V_π), all momenta outgoing (the routine flips the signs of the photon and pion momenta in place), $\mu^2 = s$, UV and IR poles dropped. The pentagons and boxes are UV-finite; their IR poles are matched by the $e\pi$ soft integrals of `NLOptions` (*inferred*).
- Normalisation: with the common factor written out, V_π carries $32\pi^2\alpha^4$ and the pentagons $\frac{(4\pi\alpha)^3}{4} \frac{\alpha}{4\pi} 2 = 8\pi^2\alpha^4$. We did not check whether the factor 4 is compensated inside the routine; the per-point comparison with CEEX agrees with CEEX.

NLOptions: last term of Equation (1) (`hard_soft_part/one_photon.f`, `soft_integrals.f`). Starting from the definition,

$$\text{NLOptions} = \frac{(4\pi\alpha)^3}{4} \text{ampNLO} \times (\text{phase space}),$$

$$\text{ampNLO} = A_{\text{LO}} \times 2[S_{jj}^{ee} - S_{12}^{ee} + S_{jj}^{\pi\pi} - S_{12}^{\pi\pi} + S^{e\pi}]$$

↓ A_{LO} (*amp_fullLO*) is the full tree, recomputed with the same helicity sums as *ampLO*

$$= |I + F|^2 \times 2[S_{jj}^{ee} - S_{12}^{ee} + S_{jj}^{\pi\pi} - S_{12}^{\pi\pi} + S^{e\pi}]$$

↓ the bracket is the eikonal factor: every pair of charged legs i, j , with the sign of $\eta_i Q_i \eta_j Q_j$, integrated over $E_\gamma < w\sqrt{s}$

$$= |I + F|^2 [\mathcal{S}_{ee}(w) + \mathcal{S}_{\pi\pi}(w) + \mathcal{S}_{e\pi}(w)] = |I + F|^2 \mathcal{S}(w) \\ = [\text{last term of Equation (1)}].$$

The ingredients ($p_1 = e^+$, $p_2 = e^-$, $q_1 = \pi^+$, $q_2 = \pi^-$):

- $S_{ij} = -\frac{\alpha}{\pi} (p_i \cdot p_j) L_{ij}$, with L_{ij} the integral of the eikonal $1/[(p_i \cdot k)(p_j \cdot k)]$ over the soft region, 't Hooft–Veltman eq. (7.3); evaluated on the momenta of the event.
- ee : *pjpj* (the e^+ self term; the factor 2 counts e^- too, the beams being symmetric) and *p1p2* (the e^+e^- term).
- $\pi\pi$: *qjqj_sub* (the π^+ and π^- self terms averaged, times 2) and *q1q2_sub* (the $\pi^+\pi^-$ term).
- $e\pi$: $S^{e\pi} = -S(p_1, q_1) - S(p_2, q_2) + S(p_1, q_2) + S(p_2, q_1)$ (*softintegral_interference*), with the signs of the charge products. $\mathcal{S}_{e\pi}$ is odd in Q_π : it makes the soft term charge-odd from $|I|^2 + |F|^2$, while $\mathcal{S}_{ee}$ and $\mathcal{S}_{\pi\pi}$ do so from $2\text{Re}[I^*F]$.
- IR regulator: $\lambda = \sqrt{s}$ for the ee and $e\pi$ terms. The two $\pi\pi$ integrals carry extra $-\frac{1}{2} \ln(q^2/s)$ terms, which amount to $\lambda^2 = q^2$ (*inferred*), the scale of the pion-vertex loops. The $\ln(2w\sqrt{s}/\lambda)$ terms carry all the w dependence of the one-photon weight.

3.3 UV renormalisation

- **Analytic pieces.** *Bornvirtualsoft*, v_s and v_f (*vertL0*) are closed-form, on-shell renormalised results: electron vertex at s and point-like pion form factor at q^2 , normalised to the charge e at $q^2 = 0$. No counterterm appears in the code; with vacuum polarisation off, α does not run.
- **Pion-line loops** (*test_vFSR.f:740–761*). This is the only place with explicit counterterms: on-shell scalar-QED counterterms built from B_0 , C_0 and the derivative $B'_0(m_\pi^2)$ (wave-function renormalisation), multiplied by tree helicity structures (*diag1*, *diag2*) and added to the result (*counter1*, *counter3*, *counter8*, *counter95*). The crossed version has no *diag1* analogue.
- **Boxes and pentagons.** The two-photon-exchange set is UV-finite and has no counterterms. Tensor integrals are reduced with Passarino–Veltman in quad precision; scalar integrals come from *QCDLoop*.
- **Regularisation.** Dimensional regularisation with $\mu^2 = s$; the $1/\epsilon$ poles are dropped and the finite parts kept. The IR poles cancel against the soft integrals, written in the same convention with $\lambda = \sqrt{s}$. Scale shifts give $\ln(q^2/s)$ terms, e.g. the extra $-\frac{1}{2} \ln(q^2/s)$ in *qjqj_sub* and *q1q2_sub*.

3.4 IR treatment

Slicing. Soft region $E_\gamma < w\sqrt{s}$ (CM frame): eikonal approximation, integrated analytically in *NLOpions*. Hard region: exact two-photon matrix element (*gen_2ph*), with the ISR·ISR (*ma/mb*), ISR·FSR (*mapi*) and FSR·FSR (*Jplus/Jmin*) amplitudes added coherently and squared with the same T_1 – T_4 pattern.

Soft subtraction in the analytic pieces. The three analytic one-loop results (ISR in the leptonic tensor, v_s for the electron vertex, `vertL0` for the pion vertex) are “virtual + soft” formulas, each with its own analytic $\ln(2w)$ soft function. All soft emission is supplied by the eikonal integrals of `NLOpions`, which also contain the $e\pi$ terms, so the analytic soft function is removed from each of the three:

- leptonic tensor (lines 6838–6856): $L - L_{\text{LO}} S_{\text{ISR}}(w)$, with

$$S_{\text{ISR}}(w) = \frac{\alpha}{\pi} \left[-\ln(4w^2) (1 + L) - \frac{1}{2}L^2 - L - \frac{\pi^2}{3} \right], \quad L = \ln \frac{m_e^2}{s};$$

- v_s (lines 2754–2761): $v_s - S_{\text{ISR}}(w)$; the $\ln(2w)$ terms cancel exactly;
- `vertL0` (line 9795): virtual part minus the soft function of Nucl. Phys. B57 (1973) 381, eq. (29), “without terms proportional to $\text{Log}(2w)$ ”.

What must hold. After these subtractions every one-loop piece should be free of analytic w terms. All w dependence then comes from `NLOpions` and cancels against the two-photon events. A leftover analytic $\ln w$ term does not cancel, and the result depends on w .

3.5 Where the errors are

Both errors are in `ampL0`, in the two blocks of Table 2 that correct the tree interference, and both add a multiple of the tree interference (the ΔB_4 , ΔB_5 of Section 3.2):

- **B4 (H1).** The vertex factor ($v_f + v_s$) is placed inside `dreal()` after the last helicity term, so it multiplies only T_4 . T_1 – T_3 of the tree interference are added a second time with coefficient 1. T_3 is a leading helicity amplitude, so this is an $\mathcal{O}(1)$ extra charge-odd term, not an $\mathcal{O}(\alpha)$ one.
- **B5 (H2).** `NLO_ISR_no_mass_terms` returns the ISR amplitude *including* the tree, not the loop alone: its leading coefficient `V1(1)` contains the tree “1” and the analytic soft function $S_{\text{ISR}}(w)$. The “1” repeats half of B3, and $S_{\text{ISR}}(w)$ is the one analytic soft function that is never removed, which breaks the condition of Section 3.4.

Both extra terms are proportional to $\text{Re}\langle I, F \rangle$, which is charge-odd. They integrate to zero over the charge-symmetric KLOE-LA cuts, so σ and all charge-even distributions are unaffected; only the pion polar angles move. Section 6 gives the code and the fixes.

Oddities without effect here. `soft_f` in `vertL0` and `ver_f` in `ampNLO` are computed but not used. The virtual-FSR term in `virt_pions` has an explicit prefactor four times that of the pentagons ($32\pi^2\alpha^4$ vs $8\pi^2\alpha^4$); we did not check whether this is compensated inside the routine, but a per-point comparison agrees with CEEX to 2.5×10^{-5} of the tree. Some “(-1)” comments do not match the code.

4 The discrepancy

Table 3: Before the fix. PHOKHARA (unpatched) 300 seeds $\times$ 500k events; CEEX more1M, 199 tasks $\times$ 5 ranks $\times$ 1M events at $E_{\text{min}} = 10^{-5}$ GeV (987 rank outputs after discarding 8 whose total diverged).

Generator	σ [nb]	$A(\theta^+)$
CEEX ($E_{\text{min}} = 10^{-5}$ GeV)	0.268385(8)	0.23726(3)
BabaYaga	0.268389(4)	0.23724(1)
PHOKHARA unpatched, this work	0.268353(46)	0.27872(16)
PHOKHARA unpatched, earlier reference	0.268393(23)	0.27861(8)

The new PHOKHARA run reproduces the earlier reference. All generators pass the charge-conjugation mirror test $\theta^+(\theta) = \theta^-(180^\circ - \theta)$, so this is not a mislabelled charge. Figure 2 shows the θ^+ distribution and Figure 5 the ratio of every observable to PHOKHARA: only the three pion angles are off, by a smooth tilt from 0.95 to 1.28.

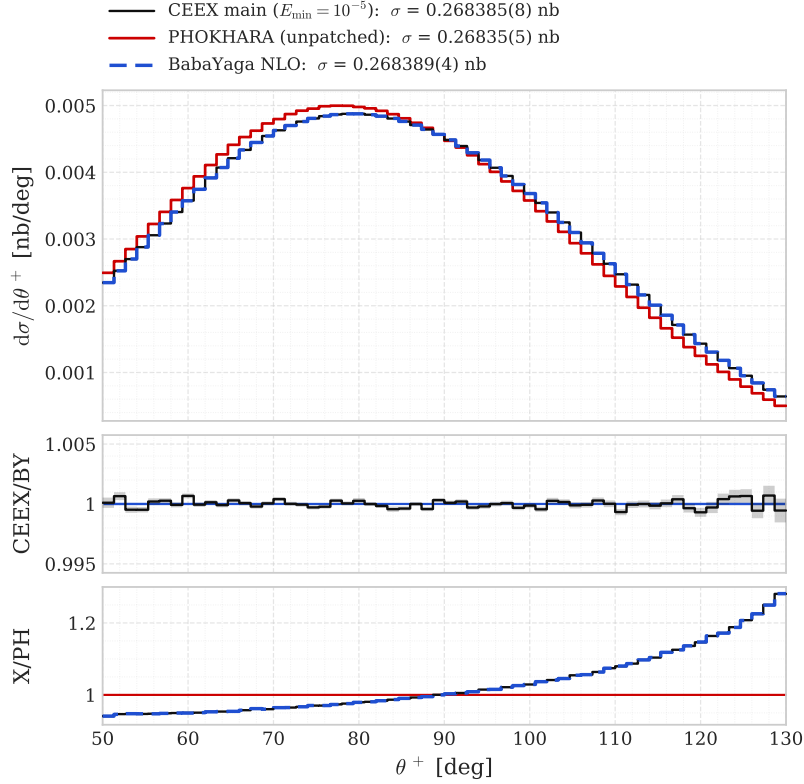


Figure 2: Before the fix. Top: $d\sigma/d\theta^+$ from CEEX ($E_{\min} = 10^{-5}$ GeV, $199 \times 5 \times 1\text{M}$, black), the unpatched PHOKHARA ($300 \times 500\text{k}$, red) and BabaYaga (blue, dashed), 60 bins; the legend gives each total cross section. Middle: CEEX/BabaYaga (CEE/BY). Bottom: CEEX and BabaYaga over PHOKHARA (X/PH). CEEX and BabaYaga agree; PHOKHARA is tilted.

At LO the codes agree exactly (earlier runs with the Strong2020 form factor: $A_{\text{LO}} = 0.0779$ and $\sigma_{\text{LO}} = 2.6523$ nb in both), so the difference lies in the NLO correction to the charge-odd part of the cross section.

5 Strategy

The difference is charge-odd (C -odd): it flips sign under $\pi^+ \leftrightarrow \pi^-$ and integrates to zero over the symmetric cuts. At $\mathcal{O}(\alpha^4)$ the C -odd part of the cross section consists of the gauge-invariant classes $Q_e^5 Q_\pi^3$ and $Q_e^3 Q_\pi^5$ (the tree interference is $Q_e^3 Q_\pi^3$). Three tools were used to isolate it:

- **Cutoff independence.** The split into one-photon (Born + virtual + soft photons below the cutoff) and two-photon (hard) parts depends on the cutoff; the sum must not.
- **Matched split.** For the same cutoff each part is uniquely defined, so the two codes must agree part by part. PHOKHARA $w = 9.804 \times 10^{-6}$ ($w\sqrt{s} = 10^{-5}$ GeV) matches CEEX $E_{\min} = 10^{-5}$ GeV.
- **Momentum swap.** At a phase-space point, the C -odd part of any piece X is $X_{\text{odd}} = \frac{1}{2}[X(p_+, p_-) - X(p_-, p_+)]$. No charge parameter is needed.

Each histogram-level test used enough statistics to resolve $A(\theta^+)$ to $\sim 10^{-3}$ while checking that σ and $M_{\pi\pi}$ are unchanged. The plan with all tests and hypotheses is in `plan.txt`.

6 The two errors in PHOKHARA

The structure of `ampL0` and the blocks B1–B5 are described in Section 3.2 and Table 2. Here I and F are the ISR and FSR tree amplitudes, v_s the electron-vertex correction at s and v_f the pion-vertex correction at q^2 .

6.1 H1: misplaced factor in the ISR×FSR correction

```

      if((nlo2.eq.1).and.(pion.eq.1))then
        ampL0 = ampL0
4      + dreal(
5        dconjg(dme*(mb(1,1)-ma(1,1))*cvac_qq)*dme*(ddpl(1,1)-ddmi(1,1))*cvac_s    ! T1
6        + dconjg(dme*(mb(2,2)-ma(2,2))*cvac_qq)*dme*(ddpl(2,2)-ddmi(2,2))*cvac_s    ! T2
7        + dconjg((-ebppb*ma(2,1)+el_m2/ebppb*mb(2,1))*cvac_qq)
8          *(-ebppb*ddmi(2,1)+el_m2/ebppb*ddpl(2,1))*cvac_s                        ! T3
9        + dconjg((ebppb*mb(1,2)-el_m2/ebppb*ma(1,2))*cvac_qq)
1       *(ebppb*ddpl(1,2)-el_m2/ebppb*ddmi(1,2))*cvac_s                        ! T4
2       *(ver_f+ver_s) )                ! multiplies T4 only

```

(Continuation lines re-indented for reading; the code is unchanged.) The block is meant to be $\text{Re}[T_1 + T_2 + T_3 + T_4](v_f + v_s)$, the amplitude-level corrections on each side of $2\text{Re}[I^*F]$. As written it is $\text{Re}[T_1 + T_2 + T_3] + \text{Re}[T_4](v_f + v_s)$, so $\text{Re}[T_1 + T_2 + T_3]$ (about a third of the tree interference) is added with coefficient 1. The muon version of the same block (`phokhara_10.0.f:10384-10394`) is correct: `2*dreal(sum)*(ver_s+ver_f)/2`.

Fix (`variants/h1.patch`): `*cvac_s *(ver_f+ver_s) ) → *cvac_s )*(ver_f+ver_s)`.

6.2 H2: tree and analytic ISR soft function in the ISR one-loop block

Block B5 contracts the FSR tree with the ISR current at one loop:

```

ampL0=ampL0+dreal( dconjg(... ddpl, ddmi ...)*( ... apl, ami ... )    ! B5

```

The current (`hard_soft_part/ISRcurrent_no_mass_terms.f`) has the coefficient

```

V1(1)=2.d0*log(4.d0*w**2)*(log(m2)+1.d0) -4.d0*log(qq2)+3.d0*log(m2) + ...    ! 1.134
V1(1)=(1.d0-2.d0*alpha/4.d0/pi*V1(1))/Sp                                     ! 1.210

```

It therefore contains the tree (the “1”) and the analytic ISR soft function, and B5 carries an extra $\text{Re}[F^*I](1 + S_{\text{ISR}}(w))$ with

$$S_{\text{ISR}}(w) = \frac{\alpha}{\pi} \left[-\ln(4w^2)(1 + L) - \frac{1}{2}L^2 - L - \frac{\pi^2}{3} \right], \quad L = \ln \frac{m_e^2}{s}.$$

- The tree interference is already added by B3.
- The analytic ISR soft function is replaced by the eikonal in `NLOpions`; it is subtracted from v_s , v_f and the leptonic tensor, but not here.
- At $w \simeq 10^{-5}$ the two parts nearly cancel ($1 + S_{\text{ISR}} = 0.044$), but S_{ISR} changes by 0.152 per decade of w . This gives the slope of the unpatched $A(\theta^+)$: $\frac{1}{2} \cdot 0.152 \cdot A_{\text{LO}} \approx 0.019$ per decade, against 0.0156 measured.

Fix (`variants/h3.patch`): subtract $\text{Re}[\sum T](1 + S_{\text{ISR}})$ after B5, with S_{ISR} computed as in the v_s subtraction.

6.3 Origin of the errors

Both errors are in the code as released by the PHOKHARA authors, not in later modifications:

- The PHOKHARA_10.0 repository used here starts from the official release: its first commit (2aaf541) contains the tarball PH10.0.tar.gz, identical (md5 f0334656...) to the one distributed at <https://looptreeduality.csic.es/phokhara/> (version 10.0, October 2020). phokhara_10.0.f and ISRcurrent_no_mass_terms.f there are byte-identical to the released ones.
- `git blame` traces every line of the H1 block and of B5, and the two V1(1) lines, to that first commit; none of the later commits touched them.
- The previous releases on the same page (8.0, 9.0, 9.1, 9.2, 9.3) have no full-NLO pion mode at all: no `nlo2` switch, no `NLO_ISR_no_mass_terms`, `NLOpions` or `virt_pions`. Both errors therefore entered with that mode in version 10.0, whose physics is described in Phys. Rev. D100 (2019) 076004.
- A first version 10.0 (December 2018) was replaced by the October 2020 one, which fixed vacuum-polarisation omissions; it is no longer distributed and could not be checked.

7 Result with the fix

phokhara_fix is the production binary with `h1.patch` and `h3.patch`.

Table 4: Fixed PHOKHARA against the soft cutoff w . The unpatched code moves by +0.0156 per decade of w .

Sample	w	σ [nb]	$A(\theta^+)$
30 seeds $\times$ 100k	10^{-3}	0.26799(33)	0.2349(12)
	10^{-4}	0.26892(35)	0.2352(13)
	10^{-5}	0.26854(32)	0.2383(12)
	10^{-6}	0.26784(31)	0.2374(11)
300 seeds $\times$ 500k	10^{-4}	0.268345(48)	0.23682(17)
	10^{-5}	0.268351(46)	0.23757(17)
	10^{-6}	0.268354(44)	0.23730(16)
1000 seeds $\times$ 5M	10^{-5}	0.268398(8)	0.23725(3)

At high statistics the slope in w is -0.00023 ± 0.00012 per decade (unpatched: +0.0156); Table 5 gives the asymmetries.

Table 5: Pion asymmetries after the fix.

Generator	$A(\theta^+)$	$-A(\theta^-)$	$-A(\theta_{\text{avg}})$
PHOKHARA fixed, 300 $\times$ 500k, w -average	0.23724(10)	0.23722(10)	0.26651(9)
PHOKHARA fixed, 1000 $\times$ 5M, $w = 10^{-5}$	0.23725(3)	0.23728(3)	0.26651(3)
CEEX, $E_{\text{min}} = 10^{-5}$ GeV, 199 $\times$ 5 $\times$ 1M	0.23726(3)	0.23726(3)	0.26650(3)
BabaYaga	0.23724(1)	0.23727(1)	0.26650(1)

At $w = 10^{-5}$ and 10^{-6} the fixed PHOKHARA agrees with CEEX and BabaYaga within $0.3\text{--}1.5\sigma$. The $w = 10^{-4}$ sample lies 0.0007 lower. To check it, the cross section was split by photon multiplicity with the fixed code at a matched cutoff of 10^{-4} GeV (PHOKHARA $w = 9.804 \times 10^{-5}$, 100 seeds $\times$ 500k; CEEX $E_{\text{min}} = 10^{-4}$ GeV), Table 6.

Table 6: Split by photon multiplicity at a matched cutoff of 10^{-4} GeV, fixed PHOKHARA.

	σ [nb]		$A(\theta^+)$	
	CEEX	PHOKHARA	CEEX	PHOKHARA
1 photon (Born + virtual + soft)	0.09533(10)	0.09539(7)	0.1548(11)	0.1558(7)
2 photons (hard)	0.17290(23)	0.17301(5)	0.2826(13)	0.2829(2)
Sum	0.26823(25)	0.26840(8)	0.2372(9)	0.2377(3)

Both parts agree, with angle-shape $\chi^2/\text{ndf} = 0.7\text{--}1.4$, and this sample gives $A(\theta^+) = 0.2377(3)$, consistent with $w = 10^{-5}$ and 10^{-6} . A 2% change of the cutoff cannot move $A(\theta^+)$ by 0.0009, so the low $w = 10^{-4}$ value is a fluctuation of that sample (2.6σ from this one). A per-point comparison with CEEX at $w = 10^{-4}$ also leaves no systematic C -odd residual after H1+H2 (mean -7×10^{-6}).

Highest statistics. A final run of 1000 seeds $\times$ 5M events (5×10^9 events passing the cuts, $w = 10^{-5}$) gives $\sigma = 0.268398(8)$ nb and $A(\theta^+) = 0.23725(3)$, $-A(\theta^-) = 0.23728(3)$, $-A(\theta_{\text{avg}}) = 0.26651(3)$. To match this precision on the CEEX side, a further CEEX run (more1M: 5 MPI ranks $\times$ 1M events per task at $E_{\text{min}} = 10^{-5}$ GeV, seeds 101–300, independent of prod500k) was run. 199 of its 200 tasks completed; the last one was stopped. After discarding 8 rank outputs whose total diverged (`check_total.sh`), the 987 rank outputs give $\sigma = 0.268385(8)$ nb and

$$A(\theta^+) = 0.23726(3), \quad -A(\theta^-) = 0.23726(3), \quad -A(\theta_{\text{avg}}) = 0.26650(3),$$

in agreement with the fixed PHOKHARA and with BabaYaga to 3×10^{-5} in all three asymmetries. Figures 3 and 4 repeat Figures 2 and 5 with these two runs: the pion-angle shape χ^2/ndf of CEEX against the fixed PHOKHARA drops from ~ 1900 to 0.8–1.5, and against BabaYaga it is 0.9–1.2.

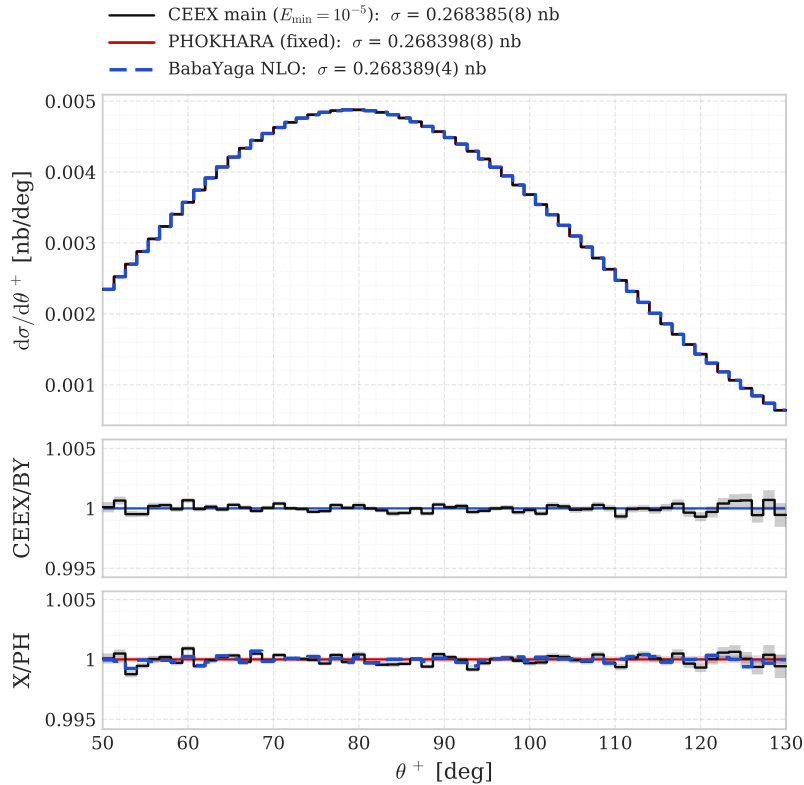


Figure 3: After the fix. As Figure 2, with the fixed PHOKHARA (1000 $\times$ 5M, $w = 10^{-5}$) in red and as the denominator of the bottom panel. The tilt is gone.

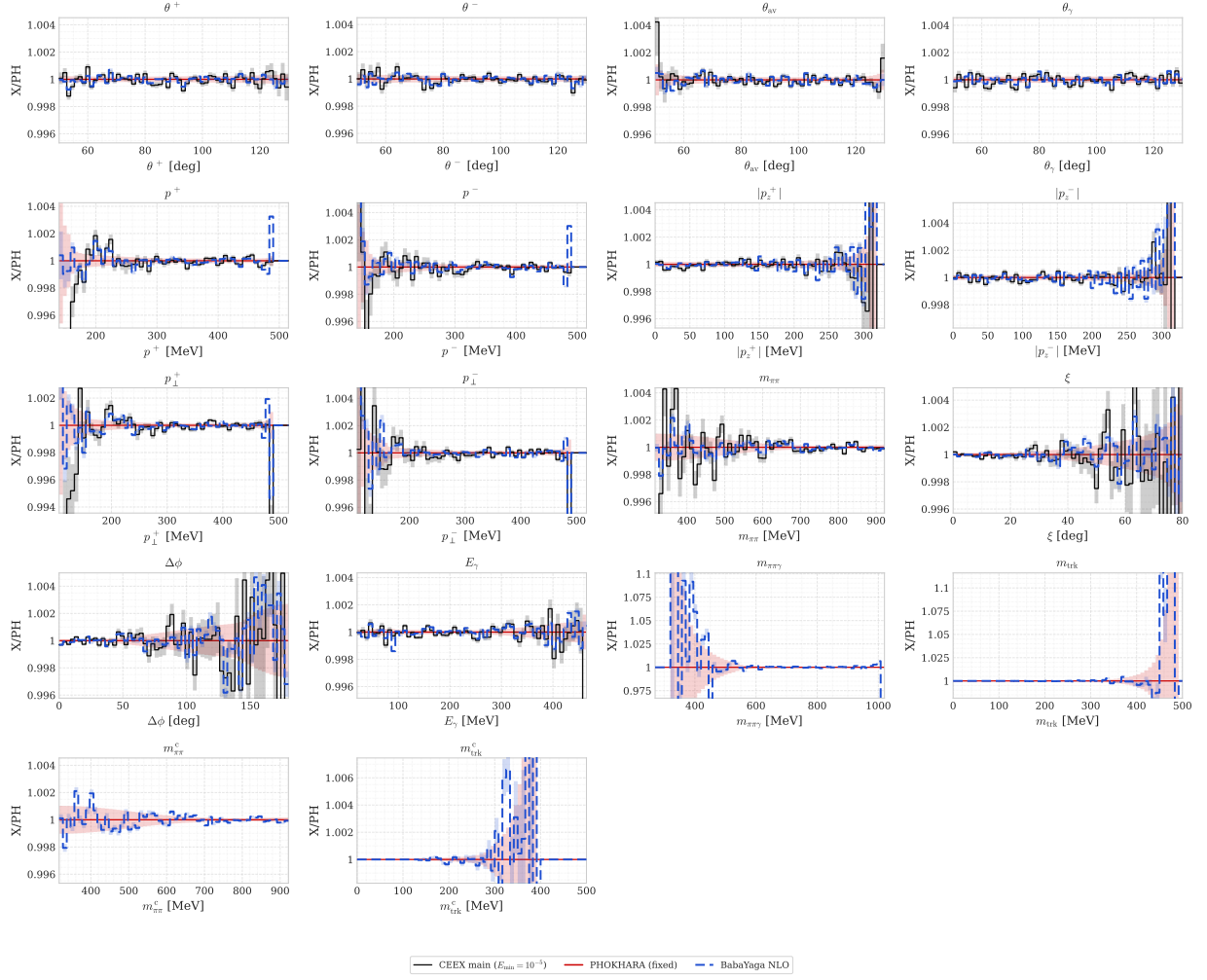


Figure 4: After the fix: every observable, CEEX ($E_{\min} = 10^{-5}$ GeV, black) and BabaYaga (blue, dashed) over the fixed PHOKHARA ($1000 \times 5M$; red band: its MC error), 60 bins. The ratio axes are clipped to the bulk of each distribution; the spikes in $m_{\pi\pi\gamma}$, m_{trk} and m_{trk}^c are nearly empty tail bins.

A Every observable before the fix

Figure 5 is the before-fix counterpart of Figure 4.

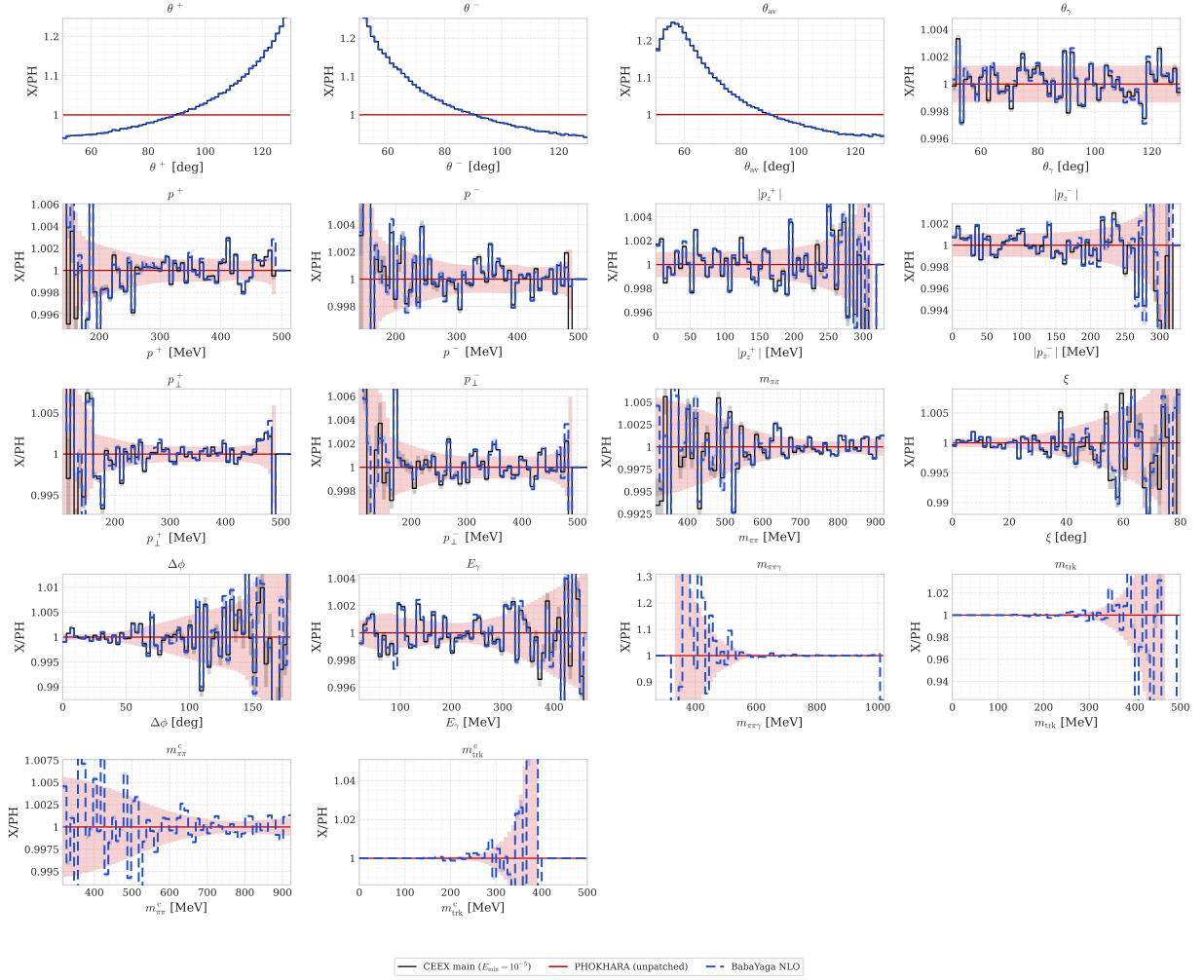


Figure 5: Before the fix: every observable, CEEEX and BabaYaga over PHOKHARA. Only θ^+ , θ^- and θ_{avg} are off; the spikes in $M_{\pi\pi\gamma}$ and M_{trk} are empty tail bins.

B Each fix alone

To show that both corrections are needed, PHOKHARA was run four times with the same card ($w = 10^{-5}$) and the same statistics (300 seeds $\times$ 500k, fresh seeds): without either fix (`phokhara_nostop`), with H1 only (`h1.patch`), with H2 only (`h3.patch`) and with both (`phokhara_fix`). Each is compared with the same CEEX run ($E_{\min} = 10^{-5}$ GeV, $199 \times 5 \times 1\text{M}$) and with BabaYaga; Table 7 gives the numbers and Figure 6 the θ^+ distributions.

Table 7: PHOKHARA with each fix alone, 300 seeds $\times$ 500k at $w = 10^{-5}$. χ^2/ndf : shape of the θ^+ distribution of CEEX and of BabaYaga against that PHOKHARA (60 bins).

PHOKHARA	σ [nb]	$A(\theta^+)$	χ^2/ndf CEEX	χ^2/ndf BabaYaga
no fix	0.268353(46)	0.27873(16)	2802.14	3146.27
H1 only	0.268428(46)	0.24224(17)	33.40	35.82
H2 only	0.268373(47)	0.27332(17)	2091.96	2339.77
H1 + H2	0.268400(46)	0.23723(17)	0.82	0.77
CEEX	0.268385(8)	0.23726(3)		
BabaYaga	0.268389(4)	0.23724(1)		

The cross section does not move, because both errors are C -odd; the asymmetry does.

- **No fix** reproduces the unpatched value of Table 3, 0.0415 (about 250σ) above CEEX.
- **H1 only** removes most of the gap, but $A(\theta^+)$ is still 0.0050 (29σ) high and the θ^+ shape is still tilted by $\pm 2\%$ ($\chi^2/\text{ndf} \approx 35$).
- **H2 only** lowers $A(\theta^+)$ by 0.0054 and leaves most of the tilt. Its size here is set by the cutoff: at $w \simeq 10^{-5}$, $1 + S_{\text{ISR}} = 0.044$ is small (Section 6.2). Without H2 the result depends on w ($+0.0156$ per decade), so no choice of w makes H1 alone correct.
- **H1 + H2** agrees with CEEX and BabaYaga within 1σ , with $\chi^2/\text{ndf} \approx 1$.

The two shifts add: -0.0365 (H1) and -0.0054 (H2), against -0.0415 for both, the same within the statistical error. This is expected, since the two corrections are separate terms in the weight.

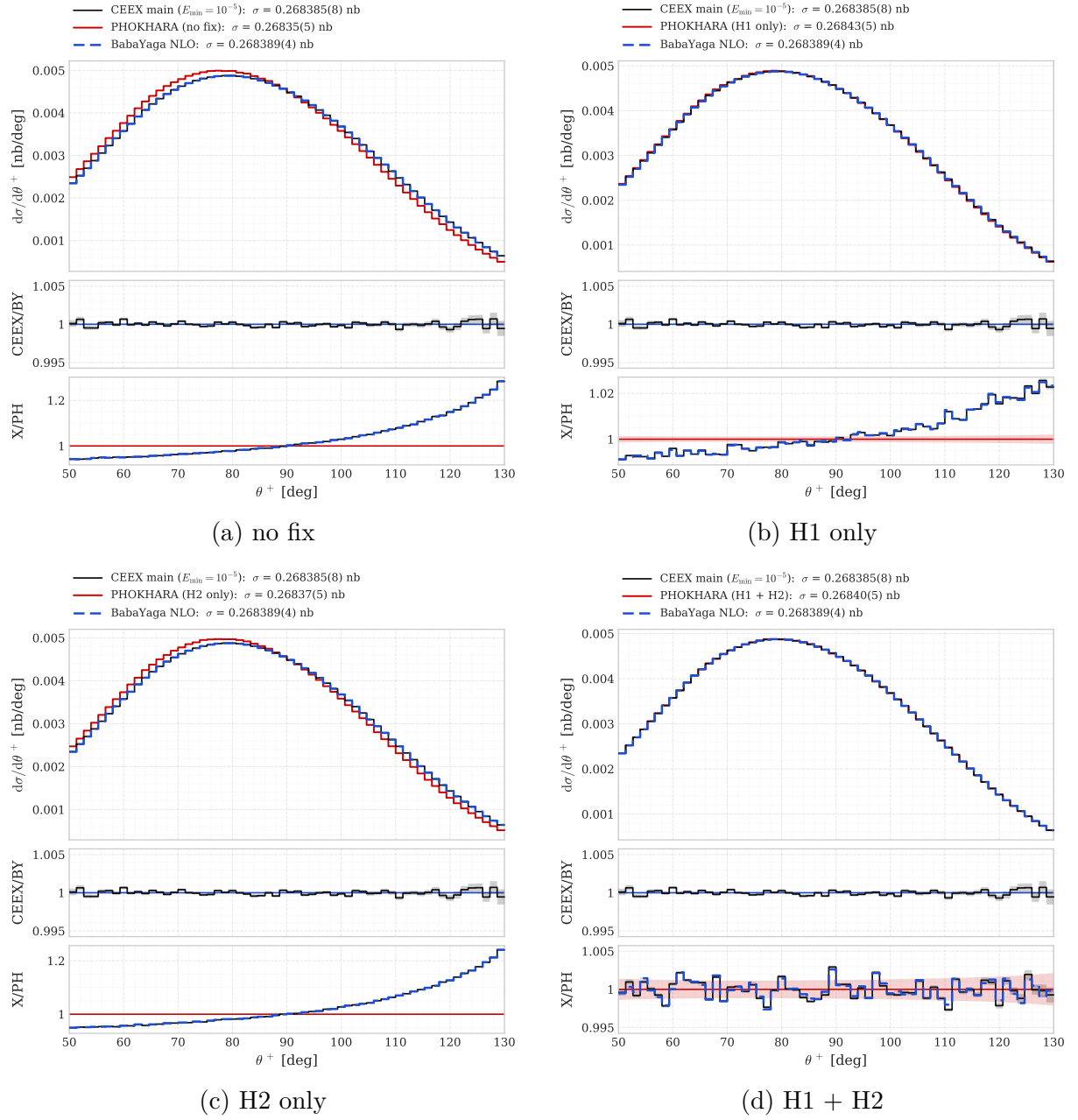


Figure 6: Each fix alone. As Figure 2, with PHOKHARA (300×500k, $w = 10^{-5}$, red) built (a) without either fix, (b) with H1 only, (c) with H2 only and (d) with both. The bottom panels (CEEX and BabaYaga over PHOKHARA) have different vertical scales. The ratio runs from 0.95 to 1.28 in (a), to 1.23 in (c) and only to 1.02 in (b); it is flat only in (d).

C Finite m_e effects

The full-NLO pion calculation is not exact in m_e everywhere. Piece by piece:

- **Exact in m_e :** the tree amplitudes I and F (massive beam spinors through the T_k), the one-loop ISR squared in `amplit` (`Bornvirtualsoft`, with its electron-mass corrections), the two-photon-exchange boxes and pentagons (electron propagators and spinors massive), and the eikonal soft integrals of `NLOpions`.
- **m_e only in logarithms:** the electron vertex v_s is the $m_e \rightarrow 0$ form, $\frac{2\alpha}{\pi}[(L-1)\ln 2w + \frac{3}{4}L - 1 + \frac{\pi^2}{6}]$ with $L = \ln(s/m_e^2)$. The neglected terms are $\mathcal{O}(m_e^2/s) \approx 2.5 \times 10^{-7}$ at fixed s , with no enhancement anywhere in phase space.
- **Massless, except logarithms and kinematics: B5,** the ISR one-loop amplitude against the FSR tree (`NLO_ISR_no_mass_terms`). Only the leading coefficient `V1(1)` contains m_e , in its soft and collinear logarithms; the other kept coefficients (`V1(2)`, `V1(5)`, `V1(8)–V1(11)`) are massless expressions in $y_{1,2} = 2p_{1,2} \cdot k/s$ and q^2/s , evaluated with the massive momenta. The seven helicity-flip coefficients (`V1(3,4,6,7,12,13,14)`) are set to zero and their expressions do not exist in the code; only their normalisation is left as a comment. The mass terms `deny1`, `deny2`, `m2miny1`, `m2miny2` are computed but not used.

So the electron mass is kept exactly in every piece except B5 (and, harmlessly, v_s). For B5 the approximation is safe at large photon angles, where the neglected terms are $\mathcal{O}(m_e^2/s)$. For a photon along a beam (KLOE small-angle) the helicity-flip terms are still suppressed, because they interfere with the helicity-flip part of F , which is itself $\propto m_e/\sqrt{s}$ with no collinear enhancement. The helicity-conserving terms of relative size $m_e^2/(p \cdot k)$ are simply missing, and matter only within $\theta_\gamma \lesssim m_e/E \approx 1$ mrad of a beam; there they are an $\mathcal{O}(\alpha)$ correction to the ISR $\times$ FSR interference, which is small for such photons.

Test at small-angle kinematics

Question. Does the massless ISR one-loop amplitude used in B5 change the one-photon matrix element when the photon is collinear with a beam? The missing mass terms can only matter within $\theta_\gamma \lesssim m_e/E \approx 1$ mrad of a beam, a region the KLOE large-angle cuts never reach but the KLOE small-angle cuts do.

Method: comparison point by point. Instead of comparing histograms, both codes evaluate the same quantity at the same phase-space points:

1. PHOKHARA generates events; at each accepted one-photon event it writes the momenta and every piece of the one-photon weight (`amplit`, the blocks B1–B5, `NLOpions`, `virt_pions`).
2. CEEX evaluates its one-photon matrix element at the same momenta: the GoSam tree and one-loop amplitudes, which keep m_e exactly, the mass counterterm, and the YFS soft factor.
3. The quantity compared is the NLO correction to the one-photon matrix element (tree + one-loop virtual + soft photons below the cutoff, minus the tree), divided by each code's own tree. No overall normalisation enters.

B5 is purely charge-odd, so we look at the charge-odd part. Every point is evaluated a second time with the π^+ and π^- momenta exchanged, and for any quantity $X_{\text{odd}} = \frac{1}{2}[X(p_+, p_-) - X(p_-, p_+)]$ and $X_{\text{even}} = \frac{1}{2}[X(p_+, p_-) + X(p_-, p_+)]$. The per-point difference is

$$D = \frac{(\delta_{\text{PH}})_{\text{odd}}}{(\text{tree}_{\text{PH}})_{\text{even}}} - \frac{(\delta_{\text{CEEX}})_{\text{odd}}}{(\text{tree}_{\text{CEEX}})_{\text{even}}},$$

with δ the NLO correction.

Removing the known errors. The PHOKHARA numbers come from the unpatched code, so D contains H1 and H2. Both are known exactly at each point from the blocks (Section 3.2): $\Delta B_4 = B_4 - \frac{1}{2}B_3(v_f + v_s)$ and $\Delta B_5 = \frac{1}{2}B_3(1 + S_{\text{ISR}}(w))$. The residual

$$R = D - \frac{(\Delta B_4 + \Delta B_5)_{\text{odd}}}{(\text{treePH})_{\text{even}}}$$

is what is left once the two errors are corrected. In the KLOE large-angle scenario R has an rms of 2.5×10^{-5} . An effect of the massless B5 would appear as a non-zero R at small photon angles.

Settings.

- **Physics:** $\sqrt{s} = 1.02$ GeV, $F_\pi = 1$, no vacuum polarisation, full NLO, soft cutoff $w\sqrt{s} = E_{\text{min}} = 10^{-5}$ GeV in both codes, the same pion mass.
- **Points:** the first 1233 accepted one-photon events (seed 77) of a PHOKHARA run with the KLOE small-angle acceptance of Strong2020 (KLOE-II): pions at 50° – 130° with $|p^z| > 90$ MeV or $p^\perp > 160$ MeV, missing momentum within 15° of a beam, $0.35 \leq M_{\pi\pi}^2 \leq 0.95$ GeV², $E_\gamma > 20$ MeV, no photon detected. Their photon angle to the nearest beam has a median of 17 mrad; 82 points lie within 1 mrad and 15 within 0.3 mrad.
- **PHOKHARA:** the unpatched code with the dump (binary `phokhara_dump_sa`).
- **CEEX:** the point driver `t4_ceex` on CEEX-main. The one parameter varied is the collinear cutoff `--thg_cutoff`. Below it CEEX does not evaluate its one-loop correction at the event: the five-point reduction becomes unreliable for nearly collinear photons (good above $\sim 3^\circ$, useless below $\sim 0.3^\circ$ according to the CEEX code). It evaluates the correction at the cutoff angle and at three times it, and extrapolates linearly in $\ln(2p \cdot k)$ to the event. We ran the default 3° (52 mrad), 1° and 0.3° .

Scale of a possible effect. In each angular bin Table 8 gives the size of the tree interference and of the whole B5 loop, $2\text{Re}[F^*\delta I_e]$ as computed by `NLO_ISR_no_mass_terms`. The error from the missing mass terms is a fraction of the B5 loop.

Table 8: Point-by-point comparison at KLOE small-angle kinematics, binned in the photon angle to the nearest beam. All entries are per point and divided by the charge-even tree. Residuals: mean over the points, with the rms in brackets, in units of 10^{-5} . The CEEX cutoff is `--thg_cutoff`; below it CEEX extrapolates its one-loop correction.

θ_γ [mrad]	0–1	1–5.2	5.2–17.5	17.5–52	52–270
Points	82	273	267	268	343
tree interference	1.9×10^{-3}	3.3×10^{-3}	1.0×10^{-2}	3.3×10^{-2}	0.12
B5 loop	3.0×10^{-4}	5.1×10^{-4}	1.6×10^{-3}	5.1×10^{-3}	1.8×10^{-2}
<i>Charge-odd residual R [10^{-5}]</i>					
CEEX cutoff 3°	+16 (140)	+7 (100)	+0.3 (48)	+0.2 (13)	−0.1 (1.0)
CEEX cutoff 1°	+5.5 (53)	+1.8 (36)	−0.0 (11)	−0.1 (1.7)	−0.1 (1.0)
CEEX cutoff 0.3°	+1.1 (8.9)	+0.2 (5.8)	−0.2 (3.6)	−0.1 (1.7)	−0.1 (1.0)
<i>Charge-even difference [10^{-5}] (not affected by B5)</i>					
CEEX cutoff 3°	+100 (56)	+59 (62)	−2 (29)	−12 (16)	−6 (4)
CEEX cutoff 1°	−73 (260)	−94 (250)	−90 (150)	−32 (47)	−6 (4)
CEEX cutoff 0.3°	−1400 (2000)	−1100 (1600)	−320 (580)	−32 (47)	−6 (4)

Results.

- **Above 3°** both codes evaluate everything directly, and the charge-odd residual is 10^{-5} of the tree, as in the large-angle scenario.

- **Below 3°** the scatter of R shrinks as the CEEX cutoff is lowered: within 1 mrad, from 1.4×10^{-3} (cutoff 3°) to 5×10^{-4} (1°) and 9×10^{-5} (0.3°). The scatter therefore comes from the CEEX extrapolation, not from PHOKHARA.
- **With the 0.3° cutoff**, R is consistent with zero in every bin. Within 1 mrad its mean is $1.1(1.0) \times 10^{-5}$ of the tree, against 3×10^{-4} for the whole B5 loop. On average, the missing mass terms change B5 by less than about 10%, i.e. by less than 3×10^{-5} of the tree.
- **The charge-even part** is not affected by B5 and is shown only to document the reference. Only the default cutoff can be used for it: at 1° and 0.3° the direct evaluation of the charge-even one-loop correction is already unstable. With the default cutoff the two codes agree to 10^{-4} above 1° and to $(0.6-1) \times 10^{-3}$ within 5 mrad, where CEEX extrapolates (quoted accuracy 5×10^{-4}). Which code is closer there cannot be decided with these tools.

The massless ISR one-loop amplitude in B5 has no visible effect, even for photons collinear to a beam. The inputs and outputs are in `RESULTS/t4_sa` (`RUN/POST_PROCESS/t4_sa_compare.py`).